

BSc (Hons) Computing - Coventry University delivered at NIBM

NB5004CEM

CYBER SECURITY

Module Prelude - Day 0

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20 Credits - Level 5

Learning Outcomes



LO1

Evaluate a range of encryption and authentication methods for a given set of requirements.

LO2

Evaluate different approaches to assessing a system's security and use these to evaluate the security of a system.

LO3

Utilise systematic knowledge to create secure environments at the host or network level.

LO4

Develop and evaluate software that addresses the most common and most severe security concerns.

Module Sessions



#	Session Topic	Layer(s)	LOs
1	Security Thinking: Extended CIA and Five-Layer Framework	<i>All</i>	LO2
2	Threat Thinking: STRIDE, ATT&CK and Trust Boundaries	<i>All</i>	LO2
3	Cryptography: Properties, Not Mechanics	<i>Data/Net/App</i>	LO1
4	Authentication: Passwords, Sessions and Tokens	<i>IAM</i>	LO1, 3
5	Authorisation, MFA and Least Privilege	<i>IAM</i>	LO1, 2
6	Application Security: OWASP Top 10 and API Risks	<i>Application</i>	LO2, 4
7	Network and System Security	<i>Net/System</i>	LO1, 3
8	Methodology, SRS Writing and Shared Responsibility	<i>All</i>	LO2
9	Assessment Tools: Scanning and Interception	<i>App/Network</i>	LO2
10	Static and Configuration Analysis	<i>App/Net/Sys</i>	LO2, 4
11	Observability, Compliance and Resilience	<i>All</i>	LO2, 3
12	Report Writing, Traceability and Viva Preparation	<i>All</i>	LO1-4

Learning Plan

12

Teaching
Sessions

3h

Per Session
theory + lab

36

Contact
Hours

200

Total Study
Hours

S1	Security Properties and Five-Layer Framework	LO2
S2	Threat Thinking: STRIDE and Trust Boundaries	LO2
S3	Cryptography: Properties Only	LO1
S4	Authentication: Passwords and Sessions	1,3
S5	Authorisation, MFA and Least Privilege	1,2
S6	Application Security: OWASP Top 10	2,4

S7	Network and System Security	1,3
S8	Methodology and SRS Writing	LO2
S9	Scanning and Interception Tools	LO2
S10	Static and Configuration Analysis	2,4
S11	Observability, Compliance and Resilience	2,3
S12	Report Writing and Viva Preparation	1-4

Learning Materials

What ships with every session.



Slide Deck

Student-facing slides for every session, available before class. Covers theory, diagrams and worked examples aligned to session learning outcomes.



Student Study Notes

Structured Markdown notes for self-directed study - one document per session. Reinforces theory and prepares you for lab exercises.



Interactive Demos - GitHub

Self-contained HTML demos you run on your own device - no install required. Distributed via the module GitHub repository and designed to work offline.



Pre-requisites

*What is expected
before Session 1.*



Networking Foundations

Basic understanding of network concepts - client-server, HTTP, ports. Deep networking knowledge is not required.



Computing Level 4

Completion of Level 4 (Year 2 equivalent) is assumed. No prior cybersecurity exposure required.



No Security Background Required

Zero prior exposure to information security is assumed. Pacing is calibrated for first-time security students.



Lab Access

A shared VM image (Kali / Ubuntu with DVWA, WebGoat, ZAP) is provided. Browser-only labs require no install.

Reference Material

ESSENTIAL - Primary

Gray Hat Hacking: The Ethical Hacker's Handbook

Harper, A., Eagle, C., Ness, J., Harris, S. et al. - McGraw-Hill Osborne, 2022 - 6th Edition

Primary reference for Sessions 6-10. Covers OWASP, penetration testing methodology, network security and exploitation techniques.

ESSENTIAL - Standards

Effective Cybersecurity: A Guide to Using Best Practices and Standards

Stallings, W. - Addison-Wesley Professional, 2018 - 1st Edition

Primary reference for Sessions 1-6 and compliance framing in Session 11. Covers CIA triad, authentication, network security and standards.

RECOMMENDED

Hacking: The Art of Exploitation

Erickson, J. - No Starch Press, 2008 - 2nd Edition

Deeper technical grounding. Excellent for students pursuing optional deeper engagement with Sessions 6-9.

Module Evaluation

You are evaluating a known system's security apparatus - not designing or building it.

PART A

30%

Context, Risks + Mini SRS

Individual report - 700 words

LO1-2

PART B

60%

Security Apparatus

Individual report - 2,000 words

LO1-4

VIVA

10%

Oral Defence

Mandatory - 7 to 10 minutes

LO1-4

Viva - Marks and Academic Integrity

The viva also verifies that Parts A and B are your own work and genuinely understood. Part A and Part B marks are capped at the level of understanding you can demonstrate in the viva - defend your work fully and your documentation marks stand in full; where you cannot demonstrate understanding of a part, the marks for that part are capped accordingly. Where the viva suggests the work may not be your own, it is referred under the academic-integrity policy.

Assessment Format

Industry Domain Assignment

Your domain is assigned by: student index mod 6

0 - Banking

1 - Healthcare

2 - Transport

3 - Education

4 - Ecommerce

5 - Hospitality

Part A - 700 words (30%)

Evaluate context, assets and roles; draw an architecture diagram with trust boundaries; identify key risks mapped to security properties; produce a mini SRS with testable acceptance criteria.

Viva - 10% (Mandatory)

7-10 minutes. Defend one design decision, present a trade-off, and handle a challenge. Marks for Parts A and B are capped at the level of understanding you can demonstrate.

Part B - 2,000 words (60%)

Evaluate the security apparatus of the known system across all five layers (IAM, Network, System, Application, Data) plus observability, resilience, compliance and a traceability matrix.

Coursework Scaffolding

Each session produces direct coursework material. Students who engage with labs leave Session 12 with a nearly-complete submission and rehearsed viva answers.

Further Points

Session Order

Sessions are cumulative - each one creates the conceptual foundation the next requires. Session 4 (Authentication) only makes sense after Session 3 (Cryptography). Attendance is sequential, not optional.

Theory Precedes Labs

The first hour of each session introduces the concept; the lab applies it. Skipping theory means the lab exercise makes no sense and produces no usable coursework output.

Terminology From Day One

Adopt correct security vocabulary from the first session. The report and viva are assessed on precision. 'The system has poor security' is not a finding. 'A07 - Identification failure violating Authenticity' is.

Approach to Evaluation

This module produces a security evaluation - not a penetration test. The goal is structured reasoning from requirements to evidence. Document your reasoning; marks are awarded for demonstrated understanding.

Background Equity

UI/UX students lead in diagramming and report writing. DevOps students lead in infrastructure. Software engineers lead in code critique. All bring value - peer activities are richer for it.

Further Points - *continued*

My Evaluation Policy

Credits for hard work. Hard means knowledge, innovation and skills - not volume.

Marks reward demonstrated understanding, correctly applied concepts and technically sound reasoning. A partially correct answer explained well is worth more than a perfect answer copied without understanding. Attempt every task - partial credit is available.

My AI-Generated Content Policy

SIMPLE. You must be able to defend your work.

Using AI tools to support your learning is permitted. Submitting AI-generated work as your own, without understanding it, is not. In the viva, you may be asked to explain any part of your submission. If you cannot explain it, you did not write it.

Use AI. Understand what it produces. Own what you submit.

Questions & Discussion

*Session 1 begins now.
Bring your curiosity - and your assigned domain.*

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